Sotto Voce: The Impacts of Technology to Enhance Worker Voice*

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Abstract

We conducted a randomized controlled trial in which we enabled workers to directly and anonymously communicate with HR through mobile-phone-based technology. Treated workers were much more likely to know about the tool than controls, though usage was low: only 5% of treated workers reported ever using it. Despite this, turnover and absenteeism were 10% and 5% lower, respectively, for treated versus control workers. Together these findings suggest a substantial option value of enhancing worker voice, which can promote positive workplace behaviors even when workers do not directly avail themselves of the technology.

Keywords: voice, exit, turnover, absenteeism, quiet quitting, productivity, readymade garments, India
JEL Classification Codes: D32, J28, M50

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1 Introduction

Relationships between managers and workers are at the heart of much of what we know about the form and function of organizations (Bandiera et al., 2007; Bloom and Van Reenen, 2007; Hoffman and Tadelis, 2021; Ichniowski et al., 1997; Lazear et al., 2015). It is not uncommon for these relationships to become strained, threatening worker wellbeing and firm growth (Akerlof et al., 2020; Coviello et al., 2022; Freeman, 1980; Freeman and Medoff, 1984; Krueger and Mas, 2004; Sandvik et al., 2021). In such instances, both parties face a choice: voice their frustrations and improve relations, or exit the employment relationship (Hirschman, 1970). Given that the costs of separation can be large for firms as well as workers (especially in high-search cost environments), the ability to communicate effectively and induce change in the employment relationship may hold substantial value for both.

Our study asks: does amplifying workers' voices – that is, enabling direct communication of grievances and concerns with employers – causally impact workplace behavior and exit decisions? We report the results of a randomized controlled trial (RCT) in which we provided access to a technology that enabled direct, anonymous communication with human resources (HR) representatives. We partnered with Labor Solutions, a technology-focused worker engagement firm, and Shahi Exports, a large garment manufacturer in India, to evaluate the impacts of a mobile phone-based worker voice tool called WOVO.

Prior to the introduction of this tool, formal channels for Shahi workers to communicate their work-related grievances were limited. WOVO allows workers to communicate anonymously with their factory's HR department via SMS. When a worker sends a message, HR representatives receive the communication via a dashboard, which they can use to sort the message, flag appropriate parties, and send responses to the worker. The key appeals of WOVO for workers are its anonymity and its ease of use. We hypothesized that enabling voice among workers might, via the logic of Hirschman (1970), allow the firm to identify workers' concerns and help resolve them, and that these improvements could reduce worker turnover and change workplace outcomes like absenteeism and productivity.

We evaluated the impacts of WOVO via a randomized encouragement design. First, the HR departments of two garment factories near Delhi were introduced to and trained in the use of

WOVO, which was completely new to the firm, in late 2018. Then, workers in the randomly assigned treatment group took part in a training session meant to familiarize them with the tool and its features through real-time demonstrations and hands-on practice. The training also stressed the anonymity of messages, which anecdotal evidence suggested was critical to workers' decisions to share frustrations with managers. We find that the training approximately doubled awareness of WOVO and increased trust in it (but not other channels of voice). Usage of WOVO also increased due to treatment – though overall engagement remained low – from essentially 0 in the control group to approximately 5 percent in the treatment group.

During the nine months while WOVO was operating, we study three important workplace behaviors using the firm's administrative data: retention, absenteeism, and productivity. Treated workers were ten percent less likely to quit ($p < 0.1$) by the end of the experiment, relative to control workers whose average attrition was about 16 percent. Enhancing voice also changed worker behavior on the intensive margin: absenteeism declined by approximately 5 percent ($p < 0.05$) relative to the control group. Finally, worker productivity (measured as the quantity of operations completed over the target number per unit time) was essentially unchanged. We also find no effects on worker salary or promotion (consistent with the absence of productivity improvements), though take-home pay increased by magnitudes consistent with the reduction in absenteeism. Retention effects disappeared and absenteeism effects became smaller after WOVO was discontinued in June 2019.

The magnitudes of the changes in workplace behavior are quite striking, given that actual use of the tool was very low. This suggests either a very large treatment on the treated (TOT) estimate of direct effects of engagement on workplace outcomes, or – perhaps more likely – that treated workers who did not directly engage with the technology were nevertheless impacted by it. This latter case would suggest that WOVO created option value. That is, even workers who did not have a specific grievance to address during the study period may have still gained utility from knowing about the existence of a reliable anonymous tool they would be able to use if the need ever arose. This could have increased retention and decreased absenteeism even for workers who did not directly avail themselves of the amenity.[1] While we do not have direct evidence of this option value mechanism,

[1] A similar result in the context of worker referrals is documented in Friebel et al. (2019): the implementation of an employee referral program in a large grocery chain increased retention even in stores where no referrals were made, most likely because workers valued being involved in the hiring process.

we show that alternative explanations (e.g., changes in supervisor behavior or changes in treated workers' attitudes for reasons unrelated to WOVO) are unlikely to be the main drivers of our results.

Our results add to the rich economics literature on the role of voice in shaping labor market dynamics. An older strand of this literature focuses on voice as represented by collective bargaining via unions (Freeman, 1980). More recent work has emphasized the endogenous emergence of voice – often through "counterproductive" workplace behavior – in response to a negative shock in the employment relationship, e.g., a pay cut (Coviello et al., 2022; Krueger and Friebel, 2022; Sandvik et al., 2021), a mass layoff (Akerlof et al., 2020), or the hiring of replacement workers (Krueger and Mas, 2004). Several studies, most closely related to our work, induce experimental variation in voice for frontline workers in low-income contexts. Adhvaryu et al. (2021) directly elicits voice using employee satisfaction surveys in India; Cai and Wang (2020) makes managers' bonuses an explicit function of worker evaluations in China; and Boudreau (2020) enforces labor regulation related to worker safety committees in Bangladesh.[2]

We build on these recent advances in two ways. First, unlike Adhvaryu et al. (2021) and Cai and Wang (2020), we enhance the worker's *ability* to voice concerns, rather than eliciting or valuing feedback directly. Conceptually, this difference helps us understand whether simply knowing that it is possible to express one's concerns to management (even when concerns come up only sporadically) can increase utility enough to affect workplace behavior. Moreover, in contrast to Adhvaryu et al. (2021), who create a worker feedback mechanism at a "special" (fraught) moment in time for workers, in which annual wage adjustments were far below expectations, we study worker voice in a business-as-usual context, increasing the generalizability of our results.

Second, while the general idea of opening a conduit for communication to upper management is similar to the worker committees studied in Boudreau (2020) (as well as Jäger et al. (2021) and Harju et al. (2021) in high-income contexts), there are two meaningful differences in the type of channel used in our study versus these papers. The aforementioned studies enable voice through committee proxy, i.e., worker representation at the board or upper management level. The technology-enabled increase in worker voice that we study prioritizes anonymity and the creation of a direct link to management. While worker representation via internal committees is typically found to have no

[2] Jäger et al. (2021) and Harju et al. (2021) study causal but non-experimental increases in worker voice at the firm level in Germany and Finland, respectively.

significant effects on retention and other workplace outcomes, we hypothesize that the novel features of this technology – anonymity and a direct link – could hold substantial value to workers, enough to create discernible changes in these outcomes.

We also contribute to the literature on managerial quality, in particular the importance of people management skills (Hoffman and Tadelis, 2021). This literature emphasizes the fact that managers' relationships with their team members are a key driver of firm success (Adhvaryu et al., 2019; Bloom et al., 2019; Bloom and Van Reenen, 2007), and demonstrates that interventions that improve managers' abilities to communicate effectively and better lead their people can generate substantial improvements in work culture and productivity (Alan et al., 2023; Bertrand and Schoar, 2003; Macchiavello et al., 2020). Our study complements this work by showing that technology designed to create an easy-to-use channel of communication between front-line workers and managers can have meaningful impacts on workplace outcomes.

2 Background

2.1 Context

Shahi is a contract manufacturer that produces apparel orders for large multinational brands. Most Shahi employees work in the sewing department, where garments are manufactured in production lines, each of which is managed by a supervisor. In addition to sewing, other departments include cutting, finishing, and other smaller departments that are not involved in the production process (e.g., maintenance or finance).

In Shahi factories, the most common way workers attempt to address workplace grievances is by directly communicating with supervisors or HR representatives. In addition, Indian law mandates the presence of worker committees, comprising of representatives of workers and the employer. Another grievance redressal mechanism is the use of suggestion boxes, where workers can submit complaints anonymously in writing (though these are rarely used in practice). The multinational brands that source from these factories also conduct independent audits (directly or through an external agency) to monitor the adherence of the factory to workplace standards.

2.2 WOVO Tool

In late 2018, Shahi launched WOVO, an SMS-based communication and grievance redressal tool, in two of its factories in Delhi National Capital Region.[3] WOVO, provided by Labor Solutions (formerly Workplace Options), is an anonymous interactive platform for SMS-based grievance redressal.[4] Each of the two factories received a unique phone number to which workers could send their suggestions or grievances. Workers bore the costs of sending SMSs.[5]

Worker-generated SMSs appear on a dashboard operated by a central HR administrator within each unit. WOVO masks the worker's phone number: only information on the department and factory floor (pre-filled from the Shahi administrative database), along with previous messages from the same number, are visible on the dashboard screen.[6] Based on the content of the SMS, the central administrator allocates the case to an HR representative. The HR representative can ask for more information from the workers through the platform via SMS. The HR representative is then responsible for entering details about the case and how it was handled in the platform. Once a case is resolved, the representative closes the case on the dashboard, informs the worker, and asks for feedback. Resolved cases are reviewed and monitored by a central team.

2.3 RCT Timeline

2.3.1 HR Training

The HR representatives were first trained to use the platform in August 2018. They were provided with guidelines about the timeline of a response, interacting with workers, and how to close a case. They were trained to ask for personal identifiers only when it was necessary to solve the grievance (in cases involving a salary error, for example). A refresher training session summarizing this content was conducted just before the launch of WOVO in November 2018.

[3]This initiative was funded by The Children's Place, a multinational children's brand.

[4]Website: https://www.laborsolutions.tech/

[5]Most phone carriers include a fixed number of free SMS messages as part of their phone plans. If not part of the bundle, messages typically cost about 1 INR (14 cents in 2019 USD).

[6]This information is only available when workers use the same number that they registered with. Almost half of the messages came from unregistered numbers, likely because it is common for workers to change numbers and/or share phones with family members.

2.3.2 Randomization

We randomized the approximately 7000 workers in two experimental factory units into a treatment and control group, stratifying by department (for non-sewing workers) or production line (for sewing department workers). Within each stratum, we randomly assigned half of the workers to treatment and the remaining to control.

2.3.3 Intervention

WOVO became operational in November 2018. It was available to all workers in the two experimental units, but we used a randomized encouragement design to generate variation in knowledge and usage. Workers in the randomly selected treatment group were trained in the tool and its features. In the first factory unit, worker training started in late November 2018, and in the second factory, training started in early December 2018. The training took place in batches – approximately two batches of 30 workers per day. During these 45-minute sessions, WOVO's anonymity and functionality were demonstrated to workers in order to build trust and familiarity.

Over the course of the trial, treatment workers were sent a total of ten reminder SMS messages.[7] These messages reminded workers about the service, encouraged them to send their suggestions and grievances, asked them to use their registered number to send messages, and provided them with information on the number of cases received and resolved. Due to technical difficulties, however, each set of SMS reminders was typically only successfully delivered to around 20–30% of intended recipients. The factory workers in our setting change phones quite frequently, which renders phone numbers in the administrative records out of date. In addition, many have registered for "do-not-disturb" services, which can lead to bulk messages being sent straight to spam. For this reason, we consider the main component of this treatment to be the initial training sessions, completed successfully in each unit.

The control group received no training or SMS messages. However, many still found out about WOVO, most likely through coworkers: at endline, around 45% of the control group reported being aware of it. Control workers had the same access to WOVO as treated workers, though nearly no one in the control group reported having used it at endline.

[7] The first SMS was sent on 15 January 2019 and approximately two messages were sent per month.

The platform generated a monthly report that included the number of cases, types of cases, average time to respond, and average time to complete a case. The median HR response time across both factories was 48 hours, while the median time to case closure was 84 hours. Appendix Table A8 provides a breakdown of the number of cases received by case type, along with examples for each type. A wide variety of cases were received, ranging from complaints about canteen food to reports of sexual harassment. The three most common categories were banking, retirement benefits scheme, and working hours/overtime. As discussed in more detail in Appendix section B, WOVO seemed useful for resolving simple cases but less effective at addressing more complicated complaints.

The funding for this initiative supported WOVO's operation until the beginning of July 2019, after which it was discontinued. The announcement regarding the closure of the tool was made through the Public Address (PA) system in the factory.

3 Data

3.1 Administrative Data

Firm administrative data provide us with information about worker tenure, basic demographics (like age and gender), as well as quit dates, attendance, productivity, and salary. We use the attendance data to calculate monthly absenteeism for each worker, from May 2018 (6 months before treatment start) to July 2019 (the month of WOVO's discontinuation). We also use monthly salary data – base monthly salary and take-home pay – for this same time period. In supplementary analysis in the online appendix, we extend the study period to the end of 2019.

We also have daily individual productivity data for production line workers. This provides us with, for each line and item on which an individual worked on a given day, a measure of productivity, defined as the number of pieces produced divided by the target number of pieces for that particular clothing item. To drop outliers that are likely the result of reporting error, we trim at the 1st and 99th percentiles of the raw productivity measure. Unlike for attendance and salary, collection of productivity data only began in late October 2018 (about one month before treatment start) and continued through July 2019.

Table 1 reports summary statistics for all workers who were at the firm when the treatment started in November 2018. There are no statistically significant differences between treatment and control workers in terms of tenure, age, gender, education, absenteeism, monthly salary, or (among those with productivity data available) productivity prior to the introduction of WOVO.

3.2 Survey Data

From the full population of workers in the two factories, we selected approximately 2600 workers to survey. Production line workers were over-sampled: 45% of all production line workers were included in the sample, compared to 30% of non-production workers. We administered a baseline survey to this sample in August 2018. The endline survey was conducted in May/June of 2019.

Both the baseline and endline surveys collected information about perceptions of HR responsiveness and awareness (which we combine into a single index and refer to as "HR responsiveness"), job satisfaction (an index combining satisfaction with the worker's current job/position, wage, supervisor, grievance redressal system, and overall workplace environment), subjective wellbeing (Cantril's ladder), and mental health (Kessler psychological distress scale).[8] In Table A1, we report summary statistics for both the administrative and survey data, restricting to workers who completed the baseline survey. The demographic and work-related characteristics of these workers are almost identical to the characteristics of the full sample (and preserve balance across treatment and control), although the surveyed workers are more likely to be female due to the oversampling of (disproportionately female) production line workers. In addition, none of the survey measures are significantly different across treatment and control. 30% of the baseline survey respondents in the control group and 29% of the treatment group did not complete the endline survey (conducted around 8 months after the baseline). Though the difference is not statistically significant, the magnitude (1.5 percentage points) is consistent with differences in retention that we document in section 4.2.

Finally, the endline survey included questions about workers' knowledge and use of the WOVO tool. The endline also included questions about how much workers trusted various people and systems to address their grievances, to which workers responded on a 5-point scale. We look separately at workers' trust in WOVO and workers' trust in all other channels (for which we average across trust measures relating to coworkers, supervisor, HR, management suggestion box, and worker committees).

[8] See Appendix section C.1 for more details.

4 Results

We begin by reporting the effects of the treatment on workers' knowledge and usage of WOVO, for which we (by design) only have endline data. We then move on to treatment effects on retention, which we estimate using a hazard model. For the remaining outcomes (absenteeism, productivity, salary, and survey outcomes) we have panel data and therefore use an ANCOVA specification (which controls for baseline values of the outcomes) to estimate treatment effects. In all regressions, the main independent variable of interest is the randomly assigned treatment indicator.

4.1 Knowledge, Use, and Trust

In Table 2, we report treatment effects on workers' knowledge and usage of WOVO, as well as trust in WOVO and other grievance redressal mechanisms. We regress these knowledge, use, and trust variables on a treatment dummy, strata fixed effects, and (only in even-numbered columns) demographic controls. Columns 1 and 2 show that workers in the treatment group are 38 percentage points more likely to report being aware of WOVO – a statistically significant and large effect relative to the control group mean of 45%. In addition to being more aware of WOVO, treatment workers are over seven times more likely ($p < 0.01$) to have used the tool (columns 3 and 4). However, tool use is quite low in *both* groups: essentially 0% in the control group and about 5% in the treatment group. Conditional on being aware of the tool, treatment workers report stronger trust in WOVO than control workers (columns 5 and 6). Trust in other grievance redressal channels is not significantly different in the treatment and control groups (columns 7 and 8).

Together these results suggest that the main "first stage" effect of the treatment was to increase worker awareness of and trust in a new technology to express grievances. Given the low usage rates, any treatment effects we find on workplace outcomes are unlikely to be driven by the actual use of WOVO and are more likely the result of option value created by its availability. Administrative data from the WOVO dashboard confirm that low self-reported usage rates were not driven by under-reporting. A total of 351 cases were submitted from 264 unique phone numbers, amounting to an average of 0.05 cases per worker and 0.04 phone numbers per worker, comparable to the 4% of workers surveyed at endline who reported having used WOVO.[9]

[9] The number of phone numbers per worker is not an exact estimate of the share of tool users in the factory because a single worker can use multiple different phone numbers.

4.2 Retention

Figure 1 separately plots cumulative quit shares among treatment and control workers, starting at the beginning of the experimental period until the end of July 2019, when the intervention ended. There is no discrete jump at the time the first SMS reminder was sent out in mid-January, consistent with the fact that only 20% of these messages were successfully delivered. About a week and a half into the experiment, the solid control line begins to separate from the dashed treatment line. The early separation of the treatment and control lines supports the idea that the mere availability (and not necessarily the use) of WOVO is what mattered – perhaps because it provided workers with the option of communicating concerns to the firm directly. The control line remains above the treatment line throughout the rest of the period, indicating that a larger share quit in the control group compared to the treatment group by the end of the study period.

The difference in the distribution of quit dates across treatment and control is not statistically significant (according to both a logrank test and a Kolmogorov-Smirnov test), but we formally test for attrition differences using a hazard model, adding controls for improved precision. Specifically, Table 3 reports the results of the following hazard regression:

$$\lambda_{ij}(t) = \lambda_0(t) \exp\left(\beta_1 T_i + \gamma X_i + \mu_j\right), \tag{1}$$

where $\lambda_{ij}(t)$ denotes the instantaneous probability of worker i in strata j quitting at time t (measured in days relative to the treatment start), conditional on being still employed at time t. T_i is an indicator variable equal to 1 if worker i belongs to the treatment group and μ_j denotes fixed effects for randomization strata (defined by unit and department or production line). We show results with and without X_i, a vector of basic demographic controls (gender, education, tenure, and age category dummies, including a dummy for those missing the education variable).

In Table 3, estimated coefficients indicate that quit rates were 6 to 10% lower in the treatment compared to the control group (hazard rates are 0.94 for column 1 and 0.90 for columns 2 and 3), implying that WOVO decreased worker exit. The estimate in column 2, which controls for tenure fixed effects in addition to age, gender, and education, is significant at the 10% level. We argue that controls for length of tenure are important because the shape of the hazard function likely varies based on the amount of time a worker has been at the firm. However, because these control variables (which were not pre-specified and instead chosen as a parsimonious set of demographic controls likely to be important predictors of our outcomes of interest) make a difference with respect to the statistical significance of our treatment effect, we also show in column 3 that our results are robust to using controls selected by the double-lasso procedure described in Urminsky et al. (2016) and Belloni et al. (2014).[10]

The effect size of 10% in this column corresponds to a treatment-control difference in quit rates of about 1 percentage point. As we discuss in Appendix section D, this magnitude is non-trivial in terms of the implied reductions in firm recruitment costs, but it is not large enough to lead to substantial differences in the composition of treatment and control workers at the end of the study period, as we discuss in section 4.4.

In Table A2, we show that coefficient magnitudes are larger for women than men, though the coefficients from the two sub-sample regressions and the difference between the two coefficients are statistically insignificant. As we discuss in Appendix section E, the retention effects do not persist after the tool is discontinued.

4.3 Absenteeism and Productivity

Next, we estimate treatment effects on worker absenteeism using the following ANCOVA specification:

$$A_{ijt} = \beta_1 \bar{A}_{i0} + \beta_2 T_i + \mu_j + \delta_t + \epsilon_{ijt}. \tag{2}$$

A_{ijt} is the share of days worker i (in strata j) was absent in month-year t and $\bar{A}_{i0}$ is average monthly absenteeism for worker i across the six months before the rollout of WOVO. We control for strata and month-year fixed effects (adding demographic controls in even-numbered columns) and cluster standard errors at the worker level. Attendance records are only available for workers still employed at the firm. In order to deal with potentially differential attrition rates by treatment status, we use a similar dynamic inverse probability weighting procedure as used in Adhvaryu et al. (2018), described in Appendix section C.3.

The first two columns of Table 4, which report the results for regression (2), show that absenteeism

[10] Details described in Appendix section C.2.

during the experimental period is significantly lower for workers in the treatment group. The effect size is 0.44-0.46 percentage points, which represents about a 5% reduction in absenteeism relative to the control group mean. The results appear to be driven by women, though our statistical power is limited. In Table A3, the effect size for women is 0.53-0.54 percentage points (significant at the 10% level). This is 60% larger than (though not significantly different from) the statistically insignificant effect for men. When we extend our analysis to the end of 2019 (Appendix section E), we find these absenteeism effects become smaller after the tool is discontinued.

Next, we study impacts on productivity, using an ANCOVA specification analogous to the one in equation (2). In this regression, a unique observation is defined by a worker, production line, item, and date, since workers sometimes work on more than one line or more than one item on the same day. In addition to strata and month-year fixed effects, we now include item fixed effects, line fixed effects, day of week fixed effects, and line item days (the number of days the line has been working on the particular item) to absorb incidental temporal and team variation in productivity. We cluster standard errors at the line level. Because productivity is missing on days a worker is not present, we use the same inverse probability weighting procedure described in Appendix section C.3.

The estimates in columns 3 and 4 of Table 4 show that treatment had no discernible effect on worker productivity. In both columns, coefficients are positive, close to zero, and statistically insignificant. Standard errors are small enough to rule out large but imprecisely estimated effects: the upper bound of the 95% confidence interval is less than 0.01 (less than 2% of control group mean). Importantly, it is not the case that production workers were less affected by the intervention in general: in Table A4, we show that the magnitudes of attrition and absenteeism effects among this sub-sample are similar to those in the full sample. We report null effects for both men and women in columns 3 and 4 of Table A3.

4.4 Salary, Promotion, and Survey Outcomes

We estimate effects on salary-related outcomes using an ANCOVA specification similar to the absenteeism regression described in equation (2). To measure the extent to which reductions in absenteeism translated into higher pay for workers, we begin by studying take-home pay (equal to base monthly salary minus insurance and retirement deductions, plus attendance bonus pay). In column 1 of Table A5, we find take-home pay increased by 0.6%. This is slightly larger than the 0.3%

increase in paid days relative to the control mean (column 2), perhaps because of attendance-related bonuses. We then move on to examine base salary, which could increase if workers receive raises in recognition for their increases in effort (evidenced by the absenteeism results). The results in column 3 reveal tightly estimated null effects.

To estimate treatment effects on a more direct measure of promotion, we use a cross-sectional regression where the dependent variable is an indicator for whether the worker was promoted between baseline and endline. Once again, column 5 reveals precise zeros. These null effects are perhaps unsurprising given that productivity did not change.

Finally, in Table A6, we show that the treatment had no effect on other worker survey outcomes: perceived responsiveness and awareness of HR and management, job satisfaction, subjective well-being, and psychological distress. In theory, because quit rates were higher in the control group, this could have led to differences in the composition of treatment and control workers by the end of the study. The weighting procedure we use alleviates concerns about differential attrition by observable characteristics. While it does not address differential attrition by unobserved characteristics, we note that the difference in quit rates was small in magnitude and unlikely to result in large compositional differences across treatment and control group. As we show in Table A7, when we restrict to workers still at the firm by the end of the study (columns 1-3), or restrict to surveyed workers who responded to the endline survey (in column 4-6) the baseline differences between treatment and control are small in magnitude and not statistically significant.

5 Alternative Explanations

We interpret our results as evidence that giving workers a means of communicating their concerns to the firm can improve retention and attendance, even if workers do not end up making use of this mechanism. However, the experiences of the treatment and control group differed in other ways – separate from their interaction with the worker voice tool – which could also be responsible for the effects we document.

5.1 Worker Attitudes Unrelated to Tool

Treated workers attended a training session that offered them some time away from their typical work activities and allowed them to interact with colleagues and superiors in a different setting. This, or the simple knowledge that one was selected for a special training program, could have improved workers' perceptions of the firm, their relationships within the firm, or satisfaction with their jobs overall. At the same time, control workers who discovered they were not chosen for this training program could have experienced the opposite effect – resulting in a gap between treatment and control driven by changes in the control group.

We argue, however, that any such effects are likely to be very small. First, workers were told that they were randomly chosen for the training as part of an experiment. This should have (but may not have completely) prevented treated workers from feeling that their selection for the training had larger implications. In addition, training or informational sessions are not out of the ordinary for Shahi workers as they are sometimes required for various brand compliance initiatives.[11] Given the short, one-time nature of the initial training, we would expect to only see effects immediately after the training. Contrary to this, in Figure 1, the treatment-control gap in retention grows over time.

If, despite all this, the training did have large and permanent effects on workers' perceptions of the firm and their career prospects, we would expect to see some differences in how workers evaluated the responsiveness of HR and management or their own overall job satisfaction in the endline survey, but we do not (columns 1 and 2 of Table A6). It seems, therefore, that the effects on retention and absenteeism cannot be solely attributed to treatment workers being more satisfied (or control workers being less satisfied) with the firm and their jobs. Moreover, the fact that the effects fade away after WOVO was discontinued (Appendix section E) suggests that the availability of the tool was a crucial driver of the effects we estimate.

The SMS reminders sent to the treatment group could have also generated an improved perception of the firm (by signalling to workers, for example, that the firm cared about their wellbeing). However,

[11]Shahi has an organizational development department that focuses primarily on worker programs. Trainings are key components of these programs and are often extensive. For example, the firm has trained 90,000 women workers on GAP inc's P.A.C.E training program which includes approximately 80 hours of training. Similarly, all production supervisors (around 2,000) undergo training through the Sustainable Textile Initiative: Together for Change program, which spans 25 hours.

as discussed above, the SMS reminders were only successfully delivered to a minority of the targeted workers each time. Moreover, there is no indication of any discrete change in retention patterns after the first SMS reminder sent in mid-January (see Figure 1).

5.2 Supervisor Behavior

If supervisors took note of who participated in the training and thought these workers were likely to report issues through the new system, they may have changed their behavior towards treatment workers in order to avoid being reported. When we examine satisfaction with one's supervisor (one component of the job satisfaction index) as its own separate outcome variable, we find no evidence of this. The estimated treatment effect is negative and small in magnitude, amounting to about 1% of a standard deviation (results available upon request). While this variable is admittedly a narrow measure of supervisor behavior, this result suggests that this channel is unlikely to be the primary driver of the results we document.

6 Discussion

This paper shows that enabling voice through a simple SMS-based tool reduces worker exit and absenteeism. We estimate that the firm's net rate of return on WOVO was 51%.[12] Given that actual usage of the tool was quite low even among treated workers, our results suggest a substantial option value of voice: simply knowing that it is possible to communicate concerns effectively, even if concerns arise only sporadically, increases worker utility enough to change workplace outcomes. It is important to note that, because of potential spillovers (through which control workers also became aware of and perhaps also increased their trust in WOVO), our estimates of the retention and absenteeism effects of the experimental treatment (the training session, essentially) are likely underestimating the effect of the availability of the worker voice tool.

From the policy perspective, our results demonstrate the promise of SMS-based tools that enhance the voice of frontline workers in low-income settings. Historically, frontline workers – particularly in settings in which collective representation is not the norm, as is the case in many low-income countries – have had very low levels of voice in the workplace. Indeed this is not a

[12]See Appendix section D for details about this calculation.

problem in low-income contexts only, as evidenced by the recent wave of so-called "quiet quitting" (low levels of workplace engagement and effort) that has plagued the US workforce. Most of the narrative around labor rights in these contexts characterizes the relationship between management and frontline workers as adversarial. The private sector is nevertheless a key player in promoting widespread access to such tools, given that firms are often the ones who bear the pecuniary and administrative costs of creating technology-enabled channels for voice. Our research shows that firms have a clear incentive to invest in amplifying workers' voices via these technologies that goes beyond public perception and marketing value; such investments have sizable economic returns by reducing costly worker turnover and absenteeism.

References

Adhvaryu, A., Kala, N., and Nyshadham, A. (2018). The skills to pay the bills: Returns to on-the-job soft skills training. Technical report, National Bureau of Economic Research.

Adhvaryu, A., Molina, T., and Nyshadham, A. (2021). Expectations, wage hikes, and worker voice. *The Economic Journal*.

Adhvaryu, A., Nyshadham, A., and Tamayo, J. A. (2019). Managerial quality and productivity dynamics. Technical report, National Bureau of Economic Research.

Akerlof, R., Ashraf, A., Macchiavello, R., and Rabbani, A. (2020). Layoffs and productivity at a bangladeshi sweater factory.

Alan, S., Corekcioglu, G., and Sutter, M. (2023). Improving workplace climate in large corporations: A clustered randomized intervention. *The Quarterly Journal of Economics*, 138(1):151–203.

Bandiera, O., Barankay, I., and Rasul, I. (2007). Incentives for managers and inequality among workers: Evidence from a firm-level experiment. *The Quarterly Journal of Economics*, 122(2):729–773.

Belloni, A., Chernozhukov, V., and Hansen, C. (2014). Inference on treatment effects after selection among high-dimensional controls. *Review of Economic Studies*, 81(2):608–650.

Bertrand, M. and Schoar, A. (2003). Managing with style: The effect of managers on firm policies. *The Quarterly journal of economics*, 118(4):1169–1208.

Bloom, N., Brynjolfsson, E., Foster, L., Jarmin, R., Patnaik, M., Saporta-Eksten, I., and Van Reenen, J. (2019). What drives differences in management practices? *American Economic Review*, 109(5):1648–83.

Bloom, N. and Van Reenen, J. (2007). Measuring and explaining management practices across firms and countries. *Quarterly Journal of Economics*, 122(4).

Boudreau, L. (2020). Multinational enforcement of labor law: Experimental evidence from bangladesh's apparel sector.

Cai, J. and Wang, S.-Y. (2020). Improving management through worker evaluations: Evidence from auto manufacturing. Working Paper 27680, National Bureau of Economic Research.

Coviello, D., Deserranno, E., and Persico, N. (2022). Counterproductive worker behavior after a pay cut. *Journal of the European Economic Association*, 20(1):222–263.

Freeman, R. B. (1980). The exit-voice tradeoff in the labor market: unionism, job tenure, quits, and separations. *The Quarterly Journal of Economics*, (June):643–674.

Freeman, R. B. and Medoff, J. L. (1984). What do unions do. *Indus. & Lab. Rel. Rev.*, 38:244.

Friebel, G., Heinz, M., Hoffman, M., and Zubanov, N. (2019). What do employee referral programs do? measuring the direct and overall effects of a management practice. Technical report, National Bureau of Economic Research.

Harju, J., Jäger, S., and Schoefer, B. (2021). Voice at work. Technical report, National Bureau of Economic Research.

Hirschman, A. O. (1970). *Exit, voice, and loyalty: Responses to decline in firms, organizations, and states*, volume 25. Harvard university press.

Hoffman, M. and Tadelis, S. (2021). People management skills, employee attrition, and manager rewards: An empirical analysis. *Journal of Political Economy*, 129(1):000–000.

Ichniowski, C., Shaw, K., and Prennushi, G. (1997). The effects of human resource management practices on productivity: a study of steel finishing lines. *The American Economic Review*, 87(3):291–313.

Jäger, S., Schoefer, B., and Heining, J. (2021). Labor in the boardroom. *The Quarterly Journal of Economics*, 136(2):669–725.

Krueger, A. B. and Mas, A. (2004). Strikes, scabs, and tread separations: labor strife and the production of defective bridgestone/firestone tires. *Journal of political Economy*, 112(2):253–289.

Krueger, M. and Friebel, G. (2022). A pay change and its long-term consequences. *Journal of Labor Economics*, 40(3):543–572.

Lazear, E. P., Shaw, K. L., and Stanton, C. T. (2015). The value of bosses. *Journal of Labor Economics*, 33(4):823–861.

Macchiavello, R., Menzel, A., Rabbani, A., and Woodruff, C. (2020). Challenges of change: An experiment promoting women to managerial roles in the bangladeshi garment sector. Technical report, National Bureau of Economic Research.

Sandvik, J., Saouma, R., Seegert, N., and Stanton, C. (2021). Employee responses to compensation changes: Evidence from a sales firm. Technical Report 12.

Urminsky, O., Hansen, C., and Chernozhukov, V. (2016). Using double-lasso regression for principled variable selection. *Available at SSRN 2733374*.

Tables and Figures

Table 1: Summary Statistics

	(1) Full sample	(2) Treatment	(3) Control	(4) Difference
Tenure (months)	48.8	49.3	48.3	-1.04
	(46.9)	(47.2)	(46.6)	(1.12)
Age (years)	34.1	34.1	34.2	0.12
	(7.77)	(7.77)	(7.77)	(0.19)
Female	0.63	0.63	0.64	0.010
	(0.48)	(0.48)	(0.48)	(0.011)
Years of Completed Schooling	8.75	8.78	8.72	-0.055
	(2.34)	(2.40)	(2.27)	(0.069)
Absenteeism	0.085	0.084	0.086	0.0022
	(0.100)	(0.098)	(0.10)	(0.0024)
Monthly Salary (rupees)	9845.0	9846.1	9843.9	-2.22
	(858.7)	(853.0)	(864.4)	(20.5)
Productivity	0.53	0.54	0.53	-0.0056
	(0.22)	(0.23)	(0.21)	(0.0090)
Observations	7046	3490	3556	7046

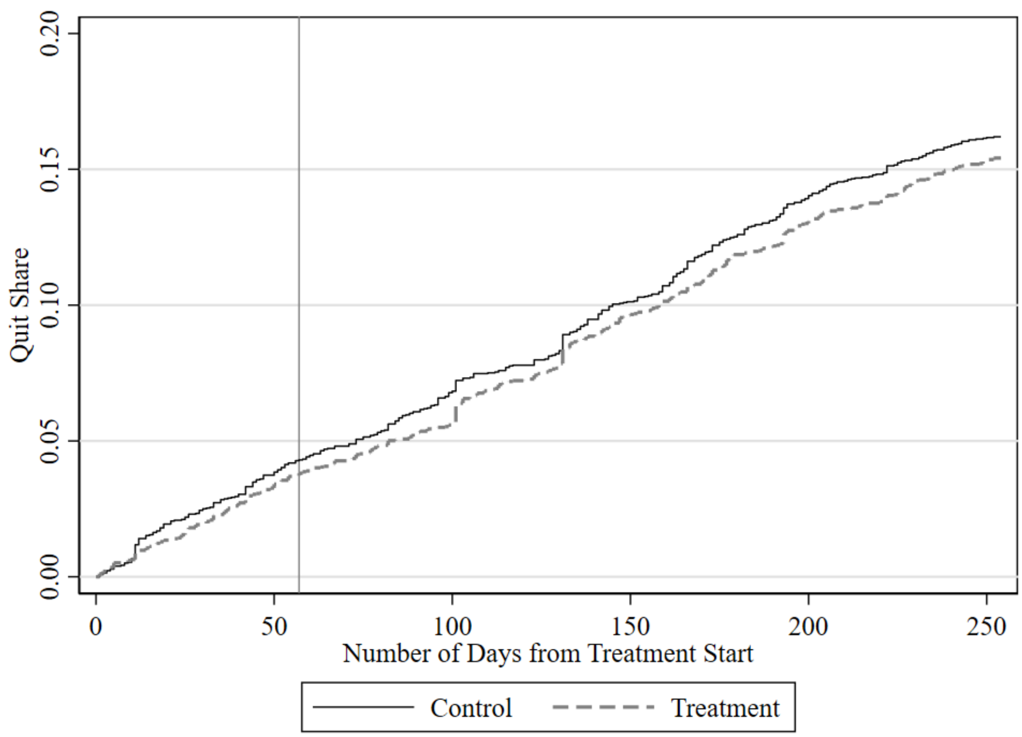
Notes: Standard deviations (in columns 1 to 3) and standard errors (in column 4) in parentheses. Includes all treatment and control workers working at the firm in November 2018. Years of education is missing for 33% of the sample. Absenteeism and salary are averaged across the 6 months prior to the tool's introduction. Productivity data, averaged across the month prior to the tool's introduction, are only available for a subset of production line workers (49% of the sample). All other variables taken from administrative data in the month that randomization took place (August 2018).

Table 2: Effect of Treatment on Tool Knowledge and Use

	(1) Aware of Tool	(2) Aware of Tool	(3) Used Tool	(4) Used Tool	(5) Trust in Tool	(6) Trust in Tool	(7) Trust in Other Channels	(8) Trust in Other Channels
Treatment	0.38***	0.38***	0.045***	0.045***	0.099*	0.091*	0.011	0.0098
	(0.021)	(0.021)	(0.0081)	(0.0080)	(0.055)	(0.055)	(0.030)	(0.030)
Observations	1831	1831	1831	1831	1119	1119	1831	1831
Control Mean	0.45	0.45	0.006	0.006	4.54	4.54	4.39	4.39
Individual Controls	No	Yes	No	Yes	No	Yes	No	Yes

Notes: Robust standard errors in parentheses. * p< 0.1, ** p< 0.05, *** p< 0.01. All regressions control for strata fixed effects. "Individual controls" include dummies for gender, age categories, tenure categories, and education categories (including a "missing" category). "Control Mean" is the mean of the dependent variable among control group workers. Observations are weighted by the inverse of the predicted probability of being in endline survey, predicted using a probit regression on treatment, demographic controls, and treatment interacted with each demographic control (tenure, age, and education categories).

Figure 1: Quit Rates by Treatment Status



Notes: Vertical line denotes approximate date of first reminder SMS.

Table 3: Effect of Treatment on Retention

	(1)	(2)	(3)
Treatment	-0.063	-0.11*	-0.10*
	(0.061)	(0.061)	(0.061)
Observations	7046	7046	7046
Individual Controls	No	Basic	Lasso

Notes: Coefficients (not hazard ratios) from a Cox proportional hazard model are reported. Robust standard errors in parentheses. * p< 0.1, ** p< 0.05, *** p< 0.01. All regressions control for strata fixed effects. "Basic" individual controls include dummies for gender, age categories, tenure categories, and education categories (including a "missing" category). "Lasso" individual controls were selected from the Belloni et al. (2014) double-lasso procedure.

Table 4: Effect of Treatment on Absenteeism and Productivity

	Absenteeism		Productivity (Quantity/Target)	
	(1)	(2)	(3)	(4)
Treatment	-0.0046**	-0.0043**	0.0025	0.0027
	(0.0021)	(0.0021)	(0.0037)	(0.0036)
Observations	51943	51943	320115	320115
Control Mean	0.096	0.096	0.51	0.51
Individual Controls	No	Yes	No	Yes

Notes: Standard errors (clustered at worker level in Columns 1-2 and at the line level in Columns 3-4) in parentheses. * p< 0.1, ** p< 0.05, *** p< 0.01. "Individual controls" include dummies for gender, age categories, tenure categories, and education categories (including a "missing" category). "Control Mean" is the mean of the dependent variable among control group workers.
Columns 1-2: The unit of observation is the worker-month, restricting to months during the experimental period. Observations are weighted by the inverse of the predicted probability of being in the sample on that month. Both regressions control for the worker's average pre-treatment absenteeism, strata fixed effects, and month-year fixed effects.
Columns 3-4: The unit of observation is the worker-day-line-item, restricting to days during the experimental period. Observations are weighted by the inverse of the predicted probability of being in the sample on that day. Both regressions control for the worker's average pre-treatment productivity, strata, month-year, day of the week, line, and item fixed effects, and line item days.

ONLINE APPENDIX

Sotto Voce: The Impacts of Technology to Enhance Worker Voice

Adhvaryu, Gade, Molina, Nyshadham

A Additional Tables and Figures

Table A1: Summary Statistics: Survey Sample

	(1) Full sample	(2) Treatment	(3) Control	(4) Difference
Tenure (months)	46.8	47.1	46.5	-0.60
	(43.8)	(42.7)	(44.9)	(1.72)
Age (years)	33.7	33.6	33.8	0.19
	(7.47)	(7.37)	(7.58)	(0.29)
Female	0.72	0.72	0.72	-0.0068
	(0.45)	(0.45)	(0.45)	(0.018)
Years of Completed Schooling	8.67	8.68	8.67	-0.018
	(2.26)	(2.23)	(2.28)	(0.11)
Absenteeism	0.073	0.073	0.073	0.00061
	(0.074)	(0.078)	(0.070)	(0.0029)
Perceived HR Responsiveness (5 pt. scale)	3.77	3.76	3.79	0.022
	(1.15)	(1.16)	(1.13)	(0.045)
Job Satisfaction (5 pt. scale)	3.91	3.89	3.92	0.027
	(0.85)	(0.85)	(0.84)	(0.033)
Subjective Wellbeing (Cantril's Ladder 10 pt. scale)	8.02	8.07	7.98	-0.090
	(2.35)	(2.33)	(2.38)	(0.092)
Kessler Psychological Distress Scale (50 pt. scale)	16.3	16.4	16.2	-0.19
	(7.77)	(7.79)	(7.76)	(0.30)
Responded to Endline Survey	0.70	0.71	0.70	-0.015
	(0.46)	(0.45)	(0.46)	(0.018)
Observations	2606	1310	1296	2606

Notes: Standard deviations (in columns 1 to 3) and standard errors (in column 4) in parentheses. Includes all treatment and control workers working at the firm in November 2018 who completed the baseline survey. Absenteeism is averaged across the 6 months prior to the tool's introduction. All other administrative variables are from the month that randomization took place (August 2018). Years of education is missing for 33% of the sample. Survey outcomes summarized here are from the baseline survey. For all variables measured on a 5-point scale, higher values represent higher satisfaction. For the Kessler Psychological Distress Scale, higher numbers represent more psychological distress. For Cantril's ladder of subjective wellbeing, a 10 represents the "best possible life."

Table A2: Effect of Treatment on Retention, by Gender

Panel A: Women	(1)	(2)
Treatment	-0.10	-0.12
	(0.079)	(0.080)
Observations	4468	4468
Individual Controls	No	Yes
Panel B: Men		
Treatment	-0.051	-0.089
	(0.10)	(0.10)
N	2578	2578
Individual Controls	No	Yes

Notes: Coefficients (not hazard ratios) from a Cox proportional hazard model are reported. Robust standard errors in parentheses. * p< 0.1, ** p< 0.05, *** p< 0.01. All regressions control for strata fixed effects. "Individual controls" include dummies for age categories, tenure categories, and education categories (including a "missing" category).

Table A3: Effect of Treatment on Absenteeism and Productivity, by Gender

	Absenteeism		Productivity (Quantity/Target)	
Panel A: Women	(1)	(2)	(3)	(4)
Treatment	-0.0054*	-0.0053*	0.0037	0.0037
	(0.0030)	(0.0030)	(0.0040)	(0.0039)
Observations	33084	33084	291467	291467
Control Mean	0.093	0.093	0.52	0.52
Individual Controls	No	Yes	No	Yes
Panel B: Men	(1)	(2)	(3)	(4)
Treatment	-0.0035	-0.0033	-0.012	-0.010
	(0.0025)	(0.0025)	(0.017)	(0.017)
N	18859	18859	28648	28648
Control Mean	0.100	0.100	0.51	0.51
Individual Controls	No	Yes	No	Yes

Notes: Standard errors (clustered at worker level in Columns 1-2 and at the line level in Columns 3-4) in parentheses. * p< 0.1, ** p< 0.05, *** p< 0.01. "Individual controls" include dummies for age categories, tenure categories, and education categories (including a "missing" category). "Control Mean" is the mean of the dependent variable among control group workers.

Columns 1-2: The unit of observation is the worker-month, restricting to months during the experimental period. Observations are weighted by the inverse of the predicted probability of being in the sample on that month. Both regressions control for the worker's average pre-treatment absenteeism, strata fixed effects, and month-year fixed effects.

Columns 3-4: The unit of observation is the worker-day-line-item, restricting to days during the experimental period. Observations are weighted by the inverse of the predicted probability of being in the sample on that day. Both regressions control for the worker's average pre-treatment productivity, strata, month-year, day of the week, line, and item fixed effects, and line item days.

Table A4: Effect of Treatment on Retention and Absenteeism, Production Sample

	Retention		Absenteeism	
	(1)	(2)	(3)	(4)
Treatment	-0.087	-0.12	-0.0058*	-0.0054*
	(0.083)	(0.084)	(0.0030)	(0.0030)
Observations	3601	3601	26458	26458
Control Mean			0.085	0.085
Individual Controls	No	Yes	No	Yes

Notes: Standard errors (clustered at worker level) in parentheses. * p< 0.1, ** p< 0.05, *** p< 0.01. "Individual controls" include dummies for gender, age categories, tenure categories, and education categories (including a "missing" category). Regressions restrict to production line workers with productivity data.
Columns 1-2: Coefficients (not hazard ratios) from a Cox proportional hazard model are reported.
Columns 3-4: The unit of observation is the worker-month, restricting to months during the experimental period. Observations are weighted by the inverse of the predicted probability of being in the sample on that month. Both regressions control for the worker's average pre-treatment absenteeism, strata fixed effects, and month-year fixed effects.

Table A5: Effect of Treatment on Salary and Promotion

	(1)	(2)	(3)	(4)
	Take-home pay (log)	Number of paid days	Base salary (log)	Promotion
Treatment	0.0063*	0.080*	0.000051	0.0075
	(0.0033)	(0.042)	(0.00025)	(0.0066)
Observations	51595	51595	51595	7046
Control Mean (in levels)	8432	29.05	10229	0.11
Control Mean (in logs)	9.01		9.23	
Individual Controls	Yes	Yes	Yes	Yes

Notes: Standard errors (clustered at worker level) in parentheses. * p< 0.1, ** p< 0.05, *** p< 0.01. "Individual controls" include dummies for gender, age categories, tenure categories, and education categories (including a "missing" category). "Control Mean" is the mean of the dependent variable among control group workers.
Columns 1-3: The unit of observation is the worker-month, restricting to months during the experimental period. Observations are weighted by the inverse of the predicted probability of being in the sample on that month, as described in Appendix section C.3. Both regressions control for the worker's average pre-treatment outcome variable, strata fixed effects, and month-year fixed effects.
Column 4: The unit of observation is the worker. The outcome variable is an indicator equal to 1 if the employee was promoted between baseline and endline.

Table A6: Effect of Treatment on Survey Outcomes

	(1) Perceived Responsiveness	(2) Job Satisfaction	(3) Subjective Wellbeing	(4) Severe Pyschological Distress	(5) Moderate Pyschological Distress
Treatment	-0.036	-0.0093	-0.11	0.0059	0.0092
	(0.034)	(0.032)	(0.097)	(0.0058)	(0.0089)
Observations	1831	1831	1831	1831	1831
Control Mean	4.36	4.21	8.52	0.01	0.04
Individual Controls	Yes	Yes	Yes	Yes	Yes

Notes: Standard errors (clustered at worker level) in parentheses. * p< 0.1, ** p< 0.05, *** p< 0.01. These worker-level regressions use endline survey measures as dependent variables and control for the same measures from the baseline survey. All regressions control for strata fixed effects. "Individual controls" include dummies for gender, age categories, tenure categories, and education categories (including a "missing" category). "Control Mean" is the mean of the dependent variable among control group workers. Observations are weighted by the inverse of the predicted probability of being in endline survey, predicted using a probit regression on treatment, demographic controls, and treatment interacted with each demographic control (tenure, age, and education categories).

Table A7: Summary Statistics: Non-Attrited Sample

	All Non-Attrited Workers			Workers in Endline Survey		
	(1)	(2)	(3)	(4)	(5)	(6)
	Treatment	Control	Difference	Treatment	Control	Difference
Tenure (months)	49.9	49.0	-0.89	48.8	47.8	-1.01
	(48.1)	(47.4)	(1.24)	(43.4)	(44.8)	(2.06)
Age (years)	34.5	34.7	0.12	34.3	34.3	0.027
	(7.74)	(7.70)	(0.20)	(7.39)	(7.47)	(0.35)
Female	0.64	0.65	0.0064	0.75	0.72	-0.024
	(0.48)	(0.48)	(0.012)	(0.44)	(0.45)	(0.021)
Years of Completed Schooling	8.71	8.74	0.027	8.59	8.69	0.11
	(2.38)	(2.22)	(0.075)	(2.13)	(2.22)	(0.12)
Absenteeism	0.080	0.081	0.00051	0.070	0.070	-0.00027
	(0.095)	(0.093)	(0.0024)	(0.075)	(0.068)	(0.0033)
Perceived HR Responsiveness (5 pt. scale)				3.75	3.76	0.0015
				(1.18)	(1.15)	(0.055)
Job Satisfaction (5 pt. scale)				3.89	3.92	0.027
				(0.86)	(0.85)	(0.040)
Subjective Wellbeing (Cantril's Ladder 10 pt. scale)				8.09	7.96	-0.13
				(2.30)	(2.40)	(0.11)
Kessler Psychological Distress Scale (50 pt. scale)				16.6	16.4	-0.22
				(7.83)	(7.84)	(0.37)
Observations	2952	2980	5932	930	901	1831

Notes: Standard deviations (in columns 1, 2, 4, 5) and standard errors (in columns 3 and 6) in parentheses. Columns 1-3 restrict to workers still working at the firm by the end of July 2019. Columns 4-6 restrict to workers who responded to the endline survey. Absenteeism is averaged across the 6 months prior to the tool's introduction. All other administrative variables are from the month that randomization took place (August 2018). Years of education is missing for 33% of the sample. Survey outcomes summarized here are from the baseline survey. For all variables measured on a 5-point scale, higher values represent higher satisfaction. For the Kessler Psychological Distress Scale, higher numbers represent more psychological distress. For Cantril's ladder of subjective wellbeing, a 10 represents the "best possible life."

B Types of Grievances

Table A8: Grievance Case Types

Case Type	Total	Examples
Banking	38	Did not receive checkbook; Need bank account number
Retirement benefit scheme	36	Retirement fund withdrawal; Check balance in the retirement fund ; Date of birth mismatch in the account
Working hours and Overtime	36	Unfair overtime or shift change; Incorrect overtime calculation
Canteen food	30	Poor food quality, taste; High price of food ; Canteen cleanliness
Welfare schemes	27	Filled the government scheme form but haven't received the payment
Compensation and Benefits	24	Received incorrect salary
Workplace conflict	19	Hostile work environment; Misbehavior; Verbal abuse
Leave	18	Difficulty in getting leaves
Exit procedures	16	Outstanding final payment after exit from the firm: Query about date of payment; Did not want to leave the firm
Maintenance	15	Machine maintainance
Factory temperature	14	Hot shop floor due to coolers not working
Housekeeping	14	Toilet closed; Toilet dirty; No soap in toilet; Shopfloor cleanliness
Sexual Harassment	8	Several cases against one person leading to investigation and action
Others	56	Test messages (Hi/Hello)

Table A8 summarizes the different types of grievances received by WOVO. Grievances related to banking were most common, followed by retirement benefit scheme, working hours and overtime, and cateen food. Though relatively uncommon, eight reports of sexual harassment were made via WOVO.

Based on information about how each case was handled, from the subset of cases that contained details about this, the tool seemed useful for resolving simple cases. For example, reports of water leakages or broken fans led to repairs and replacements. Responses to more complicated complaints, however, were likely perceived as less satisfying to the workers. One worker expressed that their salary was not enough to cover expenses, and in response, the HR administrator suggested that the worker meet with a specific person in the factory to discuss promotion opportunities and financial planning. Complaints against supervisors usually led to HR officials meeting with these supervisors, though it is not clear these meetings generated any substantial changes in supervisor behavior. Several complaints about supervisor behavior were closed with no resolution because the worker did not respond after HR asked for more information.

C Data and Empirical Strategy: Additional Details

C.1 Survey Outcomes

In the baseline and endline surveys, we asked the following questions to workers: How aware is HR/management about issues on the floor? How responsive is HR/management about issues on the floor? Workers responded using a 5-point scale, from "not responsive/aware at all" (1) to "extremely responsive/aware about all issues" (5). For each worker, we averaged the responses to these two questions for a combined measure of responsiveness and awareness, which we refer to as "HR responsiveness" in the paper.

Both surveys also included questions on job satisfaction, subjective wellbeing, and mental health. For job satisfaction, we asked workers how satisfied they were with their current job/position, current wage, supervisor, grievance redressal system, and overall workplace environment, to which they responded on a 5-point scale ranging from "extremely dissatisfied" (1) to "extremely satisfied" (5). We averaged the answers to these questions to generate a composite job satisfaction measure. We used the Kessler psychological distress scale (K10) to measure mental health. This score is generated by summing the 5-point responses to 10 different questions. This scale ranges from a minimum of 0 to a maximum of 50, with cutoff scores of 20, 25, and 30 typically used to indicate mild, moderate, and severe psychological distress, respectively. Life satisfaction is measured using Cantril's ladder, in which a response of 10 means the respondent thinks she is living her "best possible life" and a 1 corresponds to the "worst possible life."

The endline also included questions about how much workers trusted various people and systems to address their grievances. Specifically, workers were asked how much they trusted their coworkers, their supervisor, HR management, the suggestion box, worker committees, and the WOVO tool to solve their grievances. Workers responded using the following 5-point scale: "do not trust at all" (1), "do not trust very much" (2), "neither trust nor distrust" (3), "trust somewhat" (4), and "trust completely" (5). We look separately at workers' trust in WOVO and workers' trust in all other channels (for which we average across trust measures relating to coworkers, supervisor, HR, management suggestion box, and worker committees).

C.2 Double-Lasso Procedure

This section describes the double-lasso procedure used to select the controls in column 3 of Table 3. We begin with the full set of variables available in the administrative data: gender, age (continuous and in categories), tenure (continuous and in categories), skill level, education category dummies, marital status and a Hindi language dummy. We then generate all possible two-way interactions between these variables and add this to the list of possible controls. The double-lasso procedure selected the following 11 variables: 1(Tenure $>$ 7 years), Age, 1(Tenure $<$ 2 years)$\times$1(Age $<$ 25 years), 1(Tenure 2-4 years)$\times$ Female, 1(Tenure 2-4 years)$\times$1(Skill level B), 1(Tenure 2-4 years)$\times$Age, 1(Tenure 2-4 years)$\times$Age^2, 1(Tenure 2-4 years)$\times$1(Married), 1(Tenure 4-7 years)$\times$Age^2, 1(Age 35-45 years)$\times$1(Unskilled), Female$\times$1(Married).

C.3 Inverse Probability Weighting Procedure

This section describes the inverse probability weighting procedure used in the absenteeism, productivity, and salary regressions. We weight each observation using the inverse of the predicted probability of being in the sample, based on a probit regression using the following as predictors: month-year fixed effects, treatment, demographic controls (education, tenure, age), month-year fixed effects interacted with treatment and each demographic control (separately), and month-year fixed effects interacted with treatment-by-control interactions (for each demographic control).

D Rate of Return Calculation

To estimate the firm's rate of return on the tool, we combine information on the various costs associated with the intervention and our estimated treatment effects on retention and absenteeism. We consider three types of costs: subscription fees paid to the technology firm; training costs (which include productive time given up to take part in the training – for production workers and HR representatives – and the time of Shahi staff in charge of the training); and charges for reminder SMSs. Together, these yield a total cost of 16,500 in 2019 USD.[13]

Regarding benefits, the treatment reduced absenteeism by 0.43 percentage points (the more

[13] We purposely do not report costs for these three components separately given the sensitivity of information around the tool's pricing.

conservative estimate from Table 4) which translates into an additional profit of 9,820 USD.[14] In addition, the treatment increased retention by 10%, which translates into 57 fewer workers that needed to be recruited and trained (using the control group quit rate of 0.162 and treatment group sample size of 3,490). Discounted recruitment and training costs per worker are approximately 265.78 USD, which leads to a total of 15,027 USD saved due to the increase in retention.[15] Attendance and retention benefits total 24,847 USD, implying a net rate of return of approximately 51 percent.

[14]We multiply this estimate by 228 (number of working days in our sample period) and 3,490 (number of treatment workers) to obtain a total of 3,422 worker-days gained. Multiplying this by 2.87 (profit per worker per day) yields an additional profit of 9,820 USD.

[15]Recruitment costs total approximately 286 USD per worker. We use an annual interest rate of 5% and assume – conservatively – that all of the benefits are accrued at the end of the sample period, 8 months after the costs were paid.

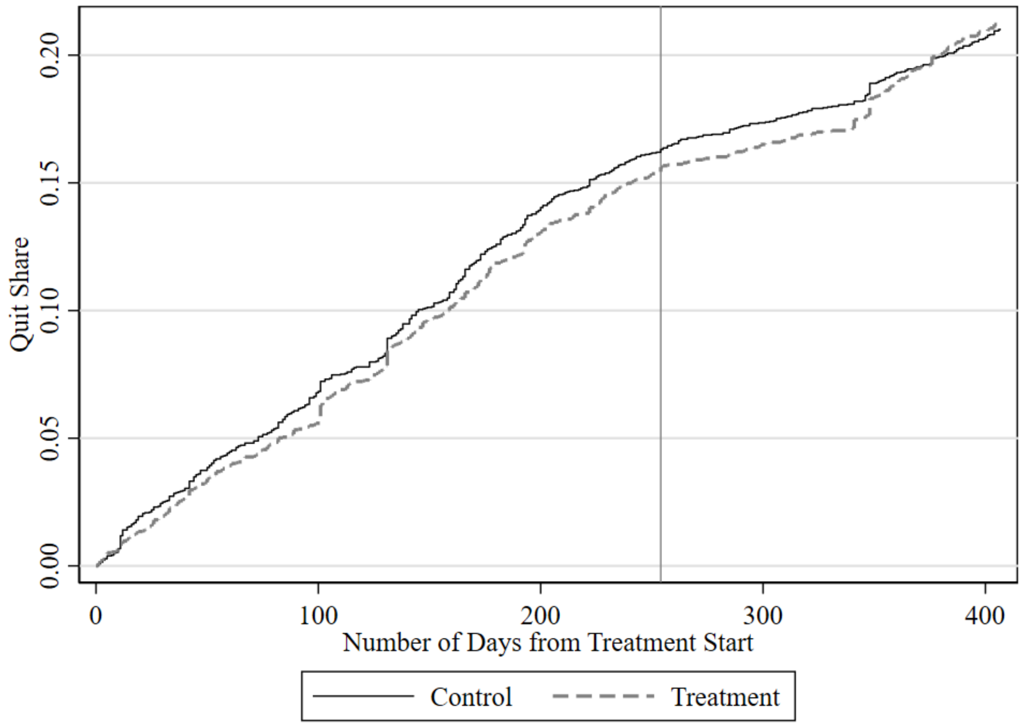
E Effects after Tool Discontinuation

In this section, we extend our study period 5 months, to the end of 2019, to examine how treatment effects on retention and absenteeism changed after the discontinuation of the tool in July 2019.[16] We begin with Figure A1, which plots cumulative quit shares in the treatment and control group until the end of December 2019. After the end of the study period in our main analysis, around 250 days after the start of the experiment, the gap between treatment and control persists for about 100 more days and then begins to close. By the end of 2019, the treatment and control lines are almost perfectly aligned (with the treatment line in fact slightly higher). The hazard regressions in the first two columns of Table A9 tell the same story. In these regressions, we create two separate time periods: our original study period (until July 31, 2019) and the remaining months of 2019. In addition to the treatment dummy, we include an interaction between treatment and a dummy for the second post-tool period. This interaction term is positive and statistically significant, reflecting the treatment group quit rates catching up to the control group, as shown in Figure A1.

In columns 3 and 4 of Table A9, we examine effects on absenteeism using the extended study period, adding an interaction between the treatment dummy and an indicator for the months after the discontinuation of the tool. The interaction term is negative but statistically insignificant – the magnitude implies a treatment effect in the months after the discontinuation that is almost one-third smaller than the treatment effect while the tool was in effect. The sum of the two coefficients, which represents the treatment effect after the tool discontinuation, is negative but statistically insignificant. In other words, the absenteeism effects do not persist after the tool is discontinued.

[16]Individual-level productivity data were not collected after July 2019.

Figure A1: Quit Rates by Treatment Status: Extended Study Period



Notes: Vertical line denotes July 31, 2019, the end of our study period in the main analysis.

Table A9: Effect of Treatment on Retention and Absenteeism: Extended Study Period

	Retention		Absenteeism	
	(1)	(2)	(3)	(4)
Treatment	-0.065	-0.11*	-0.0046**	-0.0044**
	(0.062)	(0.062)	(0.0021)	(0.0021)
Treatment x After Discontinuation	0.22**	0.23**	0.0013	0.0014
	(0.11)	(0.11)	(0.0029)	(0.0029)
Observations	13036	13036	80986	80986
Control Mean			0.092	0.092
Individual Controls	No	Yes	No	Yes

Notes: Standard errors (clustered at worker level) in parentheses. * p< 0.1, ** p< 0.05, *** p< 0.01. "Individual controls" include dummies for gender, age categories, tenure categories, and education categories (including a "missing" category).
Columns 1-2: Coefficients (not hazard ratios) from a Cox proportional hazard model are reported. Each observation is an individual-period, where the first period corresponds to the study period in the main analysis (November-July 2019), and the second period comprises the remaining months of 2019 ("After Discontinuation").
Columns 3-4: The unit of observation is the worker-month, and the time period includes both the experimental period (up to July 2019) and the remaining months of 2019 ("After Discontinuation"). Observations are weighted by the inverse of the predicted probability of being in the sample on that month. Both regressions control for the worker's average pre-treatment absenteeism, strata fixed effects, and month-year fixed effects.